Course Title & Course Code: Digital Signal Processing (7EP02)

Class: Final year Electrical **Semester**: VII

Name of the Course Teacher: Mr. P. R. Dhabe

Title of the innovative practice: Educational Video on YouTube

Objectives of the practice:

The primary objective of this innovative teaching practice is to improve conceptual clarity and self-paced learning through the use of educational videos hosted on YouTube. The specific goals are:

- To improve conceptual clarity of complex DSP topics.
- To enable flexible, self-paced learning.
- To help students visualize operations like convolution and signal manipulation.

Use of Appropriate Methods:

To achieve the above goals, the following methods were implemented:

- **Creation of Topic-Specific Educational Videos**: Videos were created covering key topics from the Power System I syllabus, including Power Circle Diagram.
- **Upload and Sharing**: Videos were uploaded to a dedicated YouTube channel and links were shared among students.
- **Interactive Content**: The videos included voice-over explanations, and problem-solving demonstrations.
- **Follow-up Sessions**: In-class discussions and Q&A sessions were conducted to clarify doubts after video viewing.

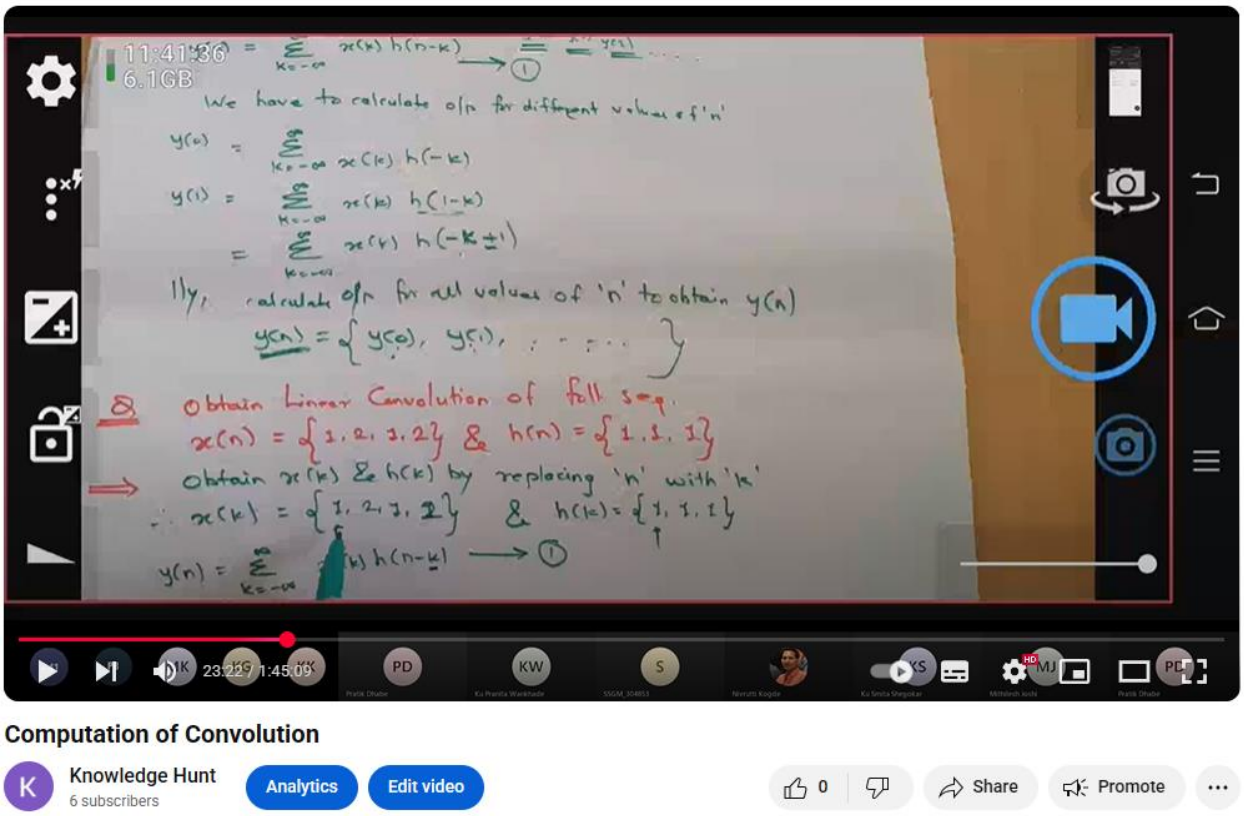
Significance of Results:

- **Improved Understanding**: Students reported better conceptual understanding of difficult topics.
- **Flexible Learning**: The videos allowed students to learn at their own pace and revisit topics as needed.
- **Extended Reach**: Videos were accessible anytime and anywhere, even beyond college hours.

Effective Presentation:

- The video content was designed using screen recording software with clear narration.
- Plots and examples shown using practical problems.
- Short links were shared during lectures for instant access.

Photos of the Activity:



Computation of Convolution

Knowledge Hunt
6 subscribers

Analytics Edit video 0 Share Promote

POs Mapped: PO1, PO2, PO4, PO12

Reflective Critique:

Faculty observed increased student interaction and curiosity during lectures that followed the video sessions. Students appreciated the convenience and clarity offered by the videos. Feedback from students included suggestions for including short quizzes at the end of each video to reinforce learning.

Evidences of Success:

- Most of the students viewed and engaged with the YouTube videos.
- Positive feedback collected indicated that most of the respondents found the video content helpful.
- Some students reported watching the videos multiple times before exams for revision.

Challenges faced during implementation:

- Initial technical difficulties in video production and editing.
- Need for subtitles or multilingual support for better inclusivity.

Peer Review & Critique Form Link: **https://forms.gle/iCgxjybVZBVbLSbh7**

Video Link: **https://youtu.be/OSJDtadnCVA**